Some days earlier, when the time difference was much less, the wormhole was not yet a time machine. It became a time machine at the first moment when something, moving at the highest possible speed, the speed of light, was able to travel along your route and arrive back at the top mouth at the very moment it started out.

If that something is a particle of light (a photon), for example, then we began with one photon and we now have two, at the starting place and time. After those two make the trip, we have four at that same place and time, then eight, then sixteen, . . . ! There is a growing crescendo of energy coursing through the wormhole, perhaps enough that the energy's gravity destroys the wormhole at the very moment it is becoming a time machine.

It would seem easy to prevent this. Just shield the wormhole from photons. However, there is something you cannot shield out: quantum fluctuations of light with ultrahigh frequency—fluctuations that inevitably exist, according to the quantum laws (Chapter 26). In 1990, Sung-Won Kim (a postdoctoral student in my research group) and I used the quantum laws to compute the fate of such fluctuations. We found a growing explosion (Figure 30.4). We thought, at first, that the explosion was too weak to destroy the wormhole. The wormhole would become a time machine despite the explosion, we thought. Stephen Hawking convinced us otherwise. The fate of the explosion is controlled by the laws of quantum gravity, he convinced us. Only when those laws are well understood will we know for sure whether backward time travel is possible.

Stephen, however, was so convinced that the ultimate answer will be no time machines, that he codified this in what he calls his "chronology protection conjecture": The laws of physics will always prevent backward time travel, thereby "keeping the universe safe for historians."